

TOTAL NON-DOMESTIC G.F.A.CALCULATION		
G/F	=	2558.753 m ²
1/F	=	1308.826 m ²
2/F	=	556.508 m ²
Grand total	=	4424.087 m ²

CALCULATION FOR NON-ESSENTIAL AREA

NON-ESSENTIAL AIR DUCTS AREA FOR 1/F	
8+9+11+12+14+15+17+18+19+20+21+22+23+24	= 75.86m ²
NON-ESSENTIAL PLANT ROOM FOR 2/F	
247.247m ² = 65.487m ² + 85.129m ² + 700.275m ² = 1,078.230m ²	
(CHILLER PUMP RM, A.H.U. RM.1, RM.2, RM.3 & RM.4, MAINTENANCE DECK)	
NON-ESSENTIAL AIR DUCTS AREA FOR 2/F	
3+4+5+6+8+9+11+12+13+14+15+17+18+19+20+21+22	= 90.09m ²

NON-ESSENTIAL PIPE DUCTS AREA FOR G/F

1+2+3+7+8+9+10+11+15+17	= 17.86m ²
NON-ESSENTIAL PIPE DUCTS AREA FOR 1/F	
1+2+3+7+8+9+10+11+15+17	= 12.94m ²

PROVISIONS OF EXIT DOORS AND EXIT ROUTES FROM A ROOM, FIRE COMPARTMENT OR STOREY											
LOCATION	INTENDED USE	NO. OF PERSON (b/m) (FROM A ROOM)	MIN. NO. OF EXIT DOOR (FROM A ROOM, FIRE COMPARTMENT OR STOREY)	MIN. TOTAL WIDTH OF EXIT DOORS (PROVIDED)	MIN. TOTAL WIDTH OF EXIT ROUTES (REQUIRED)	MIN. WIDTH OF EACH EXIT DOOR (PROVIDED)	MIN. WIDTH OF EACH EXIT ROUTE (REQUIRED)	NO. OF PERSON	MIN. NO. OF EXIT DOOR (FROM A ROOM, FIRE COMPARTMENT OR STOREY)	MIN. TOTAL WIDTH OF EXIT DOORS (PROVIDED)	MIN. TOTAL WIDTH OF EXIT ROUTES (REQUIRED)
G/F	FOYER LOUNGE	176/08/1	177	2	2	1700	4000	-	-	850	1800
	KITCHEN	25.515/4.5	8	1	1	1300	1300	750	1300	1300	1300
	TOTAL EXIT OF STOREY	177.8+8=185	2	2	2000	4700	4700	850	2000	1300	1300
	BLACK BOX CHAMBER	678.686/0.75	905	4	6	6000	9000	8900	1200	1200	1200
	FOYER	144.051/5	260	2	2	2000	2000	1000	1000	1000	1000
	BOX OFFICE	13.155/9	2	1	1	-	850	-	1000	750	800
	LAUNDRY	14.000	SAV: 3	1	1	-	850	-	1000	750	800
	CHANGING ROOM 1	7.160	SAV: 2	1	1	-	850	-	1000	750	800
	CHANGING ROOM 2	7.160	SAV: 2	1	1	-	850	-	1000	750	800
	CHANGING ROOM 3	43.159	SAV: 12	1	1	-	850	-	1000	750	800
	CHANGING ROOM 4	52.294	SAV: 12	1	1	-	850	-	1000	750	800
	CHANGING ROOM 5	8.554	SAV: 2	1	1	-	850	-	1000	750	800
	BOH AREA 2	111.119/30	4	1	1	-	850	-	1000	750	800
	TOTAL EXIT OF STOREY	905+425+2+5+2+2+12+12+12+4+1371	5	5	7500	8900	8900	1200	1200	1200	1200
1/F	MAINTENANCE GALLERY	157.967/30	6	1	1	1300	1300	750	1300	1300	1300
	FUNCTION HALL	160.310/0.75	214	2	2	3000	3000	3000	1000	1000	1000
	REHEARSAL ROOM	66.548/0.5	134	2	2	1200	1200	1200	1000	1000	1000
	TECHNICAL OFFICE & PANTRY & BREAK ROOM	139.892/8	16	-	1	-	750	-	1000	750	1000
	THEATRICAL CTRL. ROOM & STORAGE 2	44.692/30	2	-	1	-	750	-	1000	750	1000
	TOTAL EXIT OF STOREY	6+214+134+16+2=352	2	2	5	5000	5200	3000	6000	1000	1000
2/F	THEATRICAL STORAGE 3	49.337/30	2	1	1	-	750	-	1000	750	1000
	A.H.U. ROOM 1	121.763/30	5	1	1	-	750	-	1000	750	1000
	MAINTENANCE DECK	678.686/30	25	1	1	-	750	-	1000	750	1000
	TOTAL EXIT OF STOREY	2+5+25=30	1	2	-	1800	-	2100	750	900	1000

PROVISIONS OF MEANS OF ESCAPE ABOVE G/F IN CASE OF FIRE					
LOCATION AND FLOOR (COMPARTMENT)	INTENDED USE	NO. OF PERSON	U.F.A. (b/m)	FACTOR	CAPACITY OF FLOOR (COMPARTMENT)
1/F	STORAGE 3	2	49.337	49.337/30 = 1.6	3 X 1 STOREY = 3
2/F	STORAGE 3	2	49.337	49.337/30 = 1.6	3 X 1 STOREY = 3
	MAINTENANCE DECK	25	678.686	678.686/30 = 23	3 X 1 STOREY = 23
	TOTAL DISCHARGE VALUE OF STAIRCASE IN SPRINKLERED BUILDING (REQUIRED)				30
	TOTAL DISCHARGE VALUE OF STAIRCASE (SPRINKLERED BUILDING)				30
ST-1	1000 mm WIDE STAIRCASE AT U/F TO 2/F		400	400 > 342 > 30 (REQUIRED)	
ST-2	1000 mm WIDE STAIRCASE AT U/F TO 2/F		400	400 > 342 > 30 (REQUIRED)	

FIRE RESISTANCE REQUIREMENT FOR ELEMENT OF CONSTRUCTION											
LOCATION	INTENDED USE	CONSTRUCTION	USE	F.R.R. (MINUTES)	COMPARTMENT OF BUILDING	MIN. DIMENSION OF ELEMENT OF CONSTRUCTION IN mm	R.C. SLAB	R.C. BEAM	R.C. COLUMN	R.C. WALL	STEEL COLUMN
G/F	FOYER/FOYER LOUNGE	1a	4b	60			100	200	200	300	25
	BLACK BOX CHAMBER	2a	5a	60			100	200	200	300	25
	THEATRICAL B.O.H. AREAS	3	-	60			100	200	200	300	25
	STORAGE / PLANT ROOMS	4	-	60			100	200	200	300	25
1/F	FUNCTION HALL	5a	5a	60			100	200	200	300	25
	REHEARSAL ROOM	1b	5b	60			100	200	200	300	25
	VOID OF FOYER	1c	-	60			100	200	200	300	25
	VOID OF BLACK BOX CHAMBER	2b	-	60			100	200	200	300	25
2/F	THEATRICAL B.O.H. AREAS	6	-	60			100	200	200	300	25
	VOID OF PRE-FUNCTION FOYER	1c	-	60			100	200	200	300	25
	MAINTENANCE DECK (CONCRETE PORTION)	2c	-	60			100	200	200	300	25

NOTE: (1) THE SPECIAL HAZARDOUS ROOMS WALL & SLAB BY FIRE BARRIERS HAVING AN FRR OF NOT LESS THAN 150MIN.

(2) THE SPECIAL HAZARDOUS AREA IS CONNECTED DIRECTLY TO A PROTECTED EXIT, THE FIRE BARRIERS SHOULD HAVE AN FRR OF NOT LESS THAN 120MIN.

SCHEDULE OF SANITARY FITMENTS											
LOCATION	INTENDED USE	U.F.S. (b/m)	b/m PER PERSON	TOTAL NO. OF PERSON	NO. OF PERSON	MIN. NO. OF REQUIRED (UNDER PRAMP ACT 28)	W.C.	BATH	W.C.	BATH	NO. OFFERED
G/F	FOYER LOUNGE	176.086	1	177							
	BOH AREA 2	111.119	3.5	34							
	BLACK BOX CHAMBER	678.686	0.75	905							
	FOYER	144.051	5	720							
	BOX OFFICE	13.155	5	2							
	LAUNDRY	14.000	SAV: 2	2							
	CHANGING ROOM 5	8.554	SAV: 2	2							
	CHANGING ROOM 3	7.160	SAV: 2	2							
	CHANGING ROOM 4	43.159	SAV: 10	10							
	CHANGING ROOM 4	52.294	SAV: 10	10							
	CHANGING ROOM 1	7.160	SAV: 2	2							
1/F	MAINTENANCE DECK	157.968	30	214							
	FUNCTION HALL	160.310	0.75	214							
	REHEARSAL ROOM	66.548	0.5	134							
	TECHNICAL OFFICE & PANTRY & BREAK ROOM	139.892	8	16							
	THEATRICAL CTRL. ROOM & STORAGE 2	44.692	30	2							
2/F	THEATRICAL STORAGE 3	49.337	30	2							
	A.H.U. ROOM 1	121.763	30	5							
	MAINTENANCE DECK	678.686	30	25							

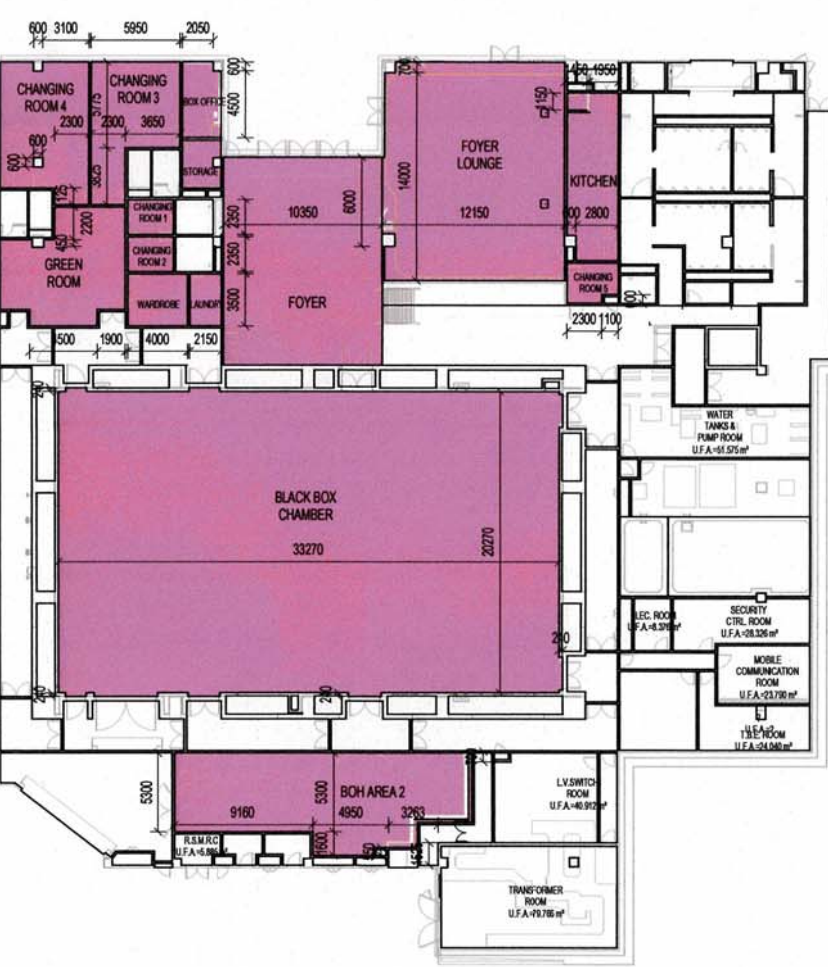


G/F G.F.A. DIAGRAM

G.F.A. CALCULATION FOR G/F

FLOOR PLATE			
DEDUCTED AREA			
1 P.D.	=	0.775 m ²	
2 P.D.	=	0.775 m ²	
3 ELEC. CABINET	=	0.698 m ²	
4 P.D.	=	0.765 m ²	
5 H.R.	=	0.475 m ²	
6 WATER METER CABINET	=	0.743 m ²	
7 P.D.	=	0.950 m ²	
8 P.D.	=	3.838 m ²	
9 P.D.	=	3.838 m ²	
10 P.D.	=	2.825 m ²	
11 P.D.	=	1.246 m ²	
12 H.R.	=	0.428 m ²	
13 H.R.	=	0.434 m ²	
14 H.R.	=	0.428 m ²	
15 H.R.	=	0.428 m ²	
16 P.D.(TEMP. CABLE PASS)	=	2.072 m ²	
17 P.D.	=	1.147 m ²	
18 H.R.	=	0.428 m ²	
19 LOADINGS/UNLOADING (ESSENTIAL PLANT RM. AREA)	=	59.474 m ²	
20 R.S.M.R.C. SPR. CTRL. VALVE RM, F.S. CTRL. CENTRE, WATER METER CABINET, ELEC. RISER RM. (ESSENTIAL PLANT RM. AREA)	=	23.226 m ²	
21 TRANSFORMER RM., L.V. SWITCH RM. (ESSENTIAL PLANT RM. AREA)	=	136.672 m ²	
22 TBE RM., FM200 CABINET, ELEC. RM. COMBINED F.S. & SPR. TANKS AND PUMP ROOM (ESSENTIAL PLANT RM. AREA)	=	222.773 m ²	
23 TOWNGAS CTRL. CABINET	=	0.375 m ²	
Grand total	=	464.810 m ²	

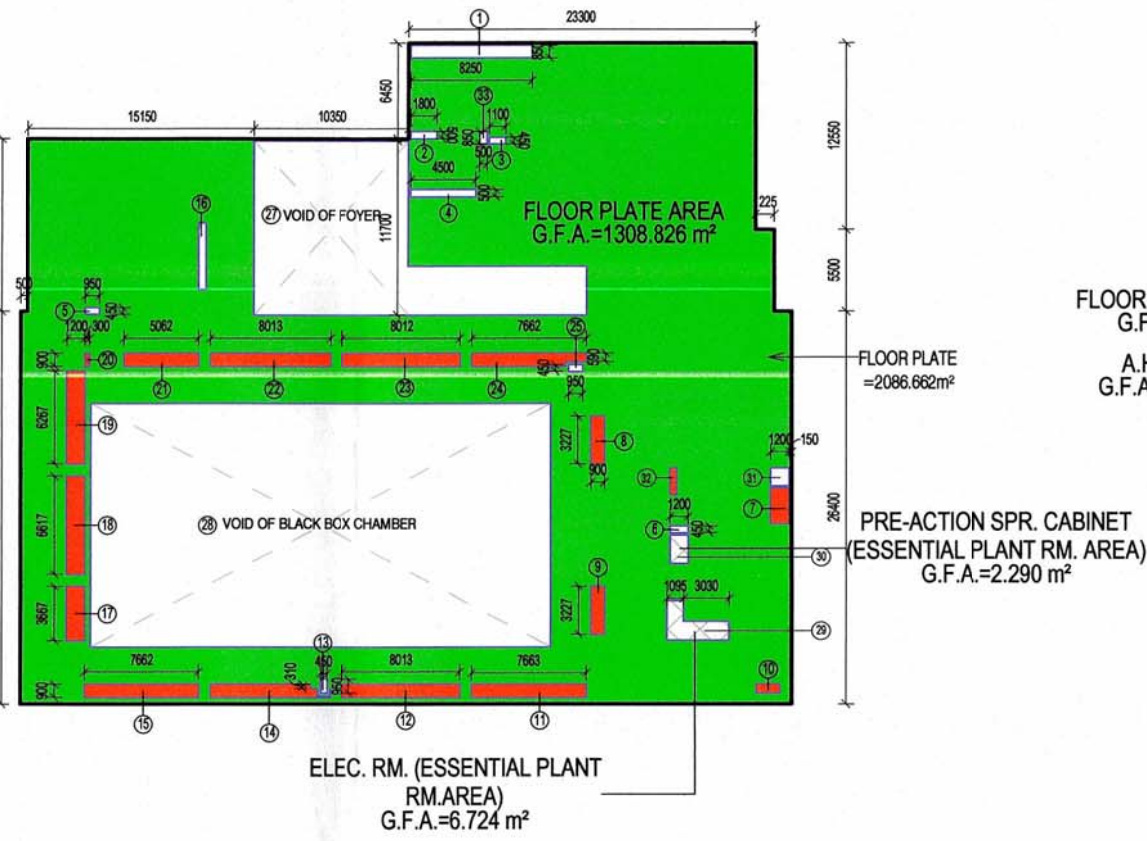
G.F.A. OF G/F	=	3023.563 sqm - 464.810 sqm	
	=	2558.753 sqm	



G/F U.F.A. DIAGRAM

G/F U.F.A. TABLE

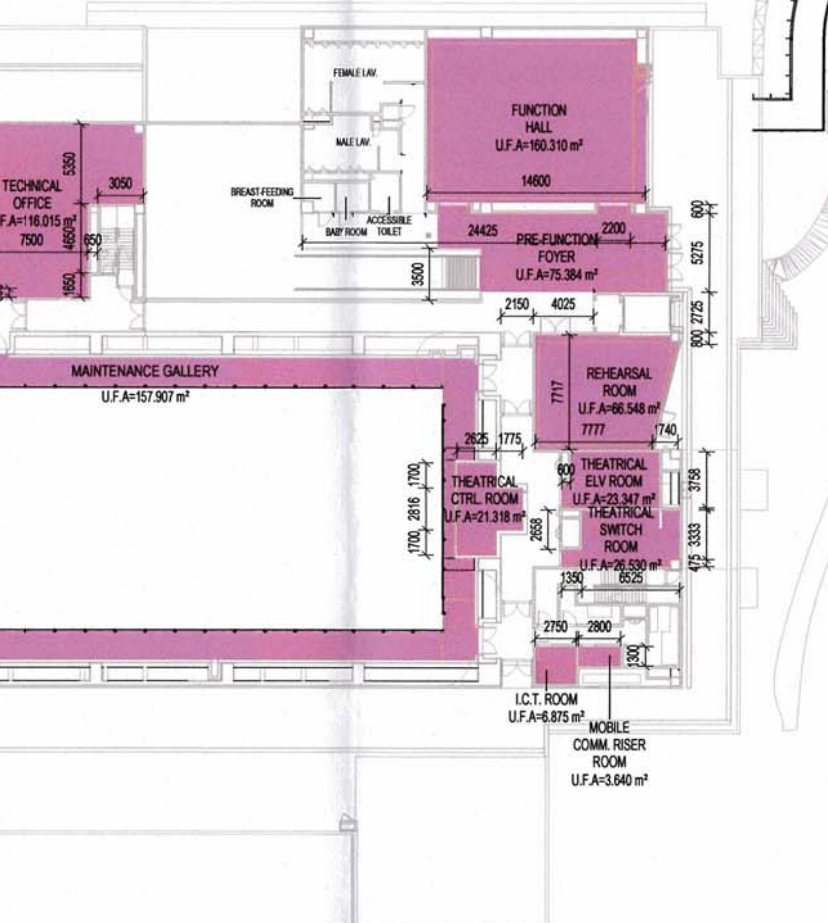
BLACK BOX CHAMBER	=	678.686 m ²
MAINTENANCE GALLERY	=	144.900 m ²
FOYER LOUNGE	=	176.086 m ²
KITCHEN	=	37.205 m ²
CHANGING ROOM 5	=	8.554 m ²
CHANGING ROOM 4	=	52.294 m ²
CHANGING ROOM 3	=	43.159 m ²
BOX OFFICE	=	13.155 m ²
WARDROBE	=	14.000 m ²
LAUNDRY	=	7.525 m ²
GREEN ROOM	=	54.505 m ²
CHANGING ROOM 1	=	7.160 m ²
CHANGING ROOM 2	=	7.160 m ²
BOH AREA 2	=	111.199 m ²
STORAGE	=	6.252 m ²
Grand total	=	1363.856 m ²



1/F G.F.A. DIAGRAM

G.F.A. CALCULATION FOR 1/F

FLOOR PLATE			
DEDUCTED AREA			
1 P.D.	=	Not Placed	
2 P.D.	=	7.013 m ²	
3 H.R.	=	0.900 m ²	
4 H.R.	=	0.485 m ²	
5 P.D.	=	2.250 m ²	
6 H.R.	=	0.427 m ²	
7 H.R.	=	0.540 m ²	
8 P.D.	=	2.860 m ²	
9 AIR DUCT	=	2.904 m ²	
10 AIR DUCT	=	0.918 m ²	
11 AIR DUCT	=	6.896 m ²	
12 AIR DUCT	=	7.211 m ²	
13 H.R.	=	0.428 m ²	
14 AIR DUCT	=	6.355 m ²	
15 AIR DUCT	=	6.896 m ²	
16 P.D.	=	2.250 m ²	
17 AIR DUCT	=	4.400 m ²	
18 AIR DUCT	=	7.940 m ²	
19 AIR DUCT	=	7.520 m ²	
20 AIR DUCT	=	0.270 m ²	
21 AIR DUCT	=	4.556 m ²	
22 AIR DUCT	=	7.211 m ²	
23 AIR DUCT	=	6.465 m ²	
24 AIR DUCT	=	0.420 m ²	
25 H.R.	=	160.850 m ²	
27 VOID OF FOYER	=	508.008 m ²	
28 VOID OF CHAMBER	=	6.724 m ²	
29 ELEC. RM.	=	2.290 m ²	
30 PRE-ACTION SPR. CABINET	=	1.450 m ²	
31 H.R.	=	0.810 m ²	
32 P.D.	=	0.425 m ²	
33 P.D.	=	777.836 m ²	
Grand total: 33	=	777.836 m ²	
G.F.A. OF 1/F	=	2086.662 sqm - 774.836 sqm	
	=	1308.826 sqm	



1/F U.F.A. TABLE

TECHNICAL OFFICE	=	116.015 m ²
MAINTENANCE GALLERY	=	157.907 m ²
THEATRICAL CTRL. ROOM	=	21.318 m ²
I.C.T. ROOM	=	6.875 m ²
THEATRICAL ELV ROOM	=	23.347 m ²
THEATRICAL SWITCH ROOM	=	26.530 m ²
REHEARSAL ROOM	=	65.548 m ²
FUNCTION HALL	=	160.310 m ²
MOBILE COMM. RISER ROOM	=	3.640 m ²
PANTRY	=	6.587 m ²
BREAK ROOM	=	15.550 m ²
PRE-FUNCTION FOYER	=	75.384 m ²
Grand total	=	681.810 m ²